CT-361 Artificial Intelligence & Expert System
Complex Computing Problem (CCP)

Course Learning Outcome:
CLO 2 (C2): Identify appropiate AI techniques to solve classification problems of moderate complexity.

Complex Computing Problem Attributes:

- **CP 2 Depth of analysis required:** Has no obvious solution, and requires conceptual thinking and innovative analysis to formulate suitable abstract models
- **CP 3 Depth of knowledge required:** A solution requires the use of in-depth computing or domain knowledge and an analytical approach that is based on well-founded principles.

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Project Title & Problem Statement:

Students must define a real-world problem within domains such as healthcare, cybersecurity, finance, environmental monitoring, marketing, or smart systems that can be addressed using AI and Expert System techniques. The project title should be clear and relevant. The problem statement must briefly explain:

- The issue being addressed
- Its real-world impact
- The AI/Expert System approach to be applied (e.g., rule-based reasoning, predictive modeling, NLP, object detection, or hybrid AI methods)

Problem Description:

The project should address a meaningful real-world problem where intelligent decision-making, prediction, or automated reasoning is required. Students must clearly describe the selected domain, the specific use case, and the goals of the system. The solution should demonstrate how AI or Expert System techniques can transform raw data or domain knowledge into accurate predictions, recommendations, or automated decisions.

At least two datasets or knowledge sources must be used and briefly described (source, type of data, and relevance). Students should explain how these will be utilized to build and evaluate the proposed AI system using suitable techniques and performance metrics.

Assumptions & Constraints:

- Projects must be completed in designated groups.
- At least two relevant datasets must be used per project.
- Datasets may contain noise, missing values, or inconsistencies; appropriate pre-processing is required.
- Systems should be able to handle diverse data types depending on the chosen domain.
- External factors (e.g., environment, user behavior, dynamic input) may influence system performance.
- All models must adhere to data privacy regulations (e.g., anonymizing personal/sensitive data).
- Ethical considerations must be followed—ensure fairness, transparency, and accountability.
- A baseline model must be implemented, with performance improvement using multiple AI approaches.
- The performance of real-time monitoring depends on network availability

Deliverables:

- **Python Code Implementation:** ***Design*** a complete AI solution for a real-world problem using at least two datasets, including preprocessing, model building, evaluation, and performance improvement (e.g., tuning or ensembling).
- **Visualization:** ***Develop*** a simple, user-friendly visualization or interface to present results. This could include charts, maps or prediction outputs. The goal is to make the findings interpretable and actionable for users.
- **Technical Report:** ***Compile*** a minimum 10-page technical report covering methodology (with diagram), models used, results, and challenges.

Complex Computing Problem Assessment Rubrics

Student Name: _____ Student Roll No.: _____

Criteria and Scales		
Satisfactory (5)	**Average (3)**	**Unsatisfactory (1)**
Criterion 1: Code: To what extent is the code complete, accurate, and robust, including handling of technical challenges and assumptions? **(CP3: *Depth of Knowledge Required*)**		
Code is complete, accurate, and well-structured. Assumptions are clear and valid. Technical issues are effectively handled.	Code is mostly correct with minor issues. Assumptions are partially justified. Some technical problems are addressed.	Code is incomplete or incorrect. Assumptions are unclear or missing. Technical challenges are poorly handled or ignored.
Criterion 2: Report: How thorough and well organized is the solution **(CP2: *Depth of Analysis Required*)**		
All the necessary information clearly organized for easy use in solving the problem	Mediocre information which may or may not lead to a solution	No report provided

Total Marks: _____/10____

Teacher's Signature:_____